AI Applications in Research

Core Concepts and the Practical Use of Claude Code

Concept summary compiled from *Scott's Mixtape Substack*

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Abstract

This document summarizes the main concepts surrounding the use of artificial-intelligence (AI) agents in academic and quantitative research, with particular attention to **Claude Code** as a research "harness." It synthesizes the recurring themes developed by economist Scott Cunningham in his newsletter *Scott's Mixtape Substack* through late 2026—agentic workflows, hooks, connectors and the Model Context Protocol (MCP), memory management, the human-in-the-loop question, and the discipline of *production, verification and understanding*. It also situates these tools within their emerging economic context (the rise of the "solopreneur" and the link between labor productivity and per-capita income). The aim is a compact, scannable reference to the ideas, not a tutorial.

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1 Introduction: agents enter the research workflow

Over 2025–2026 AI coding agents matured from autocomplete assistants into systems capable of carrying substantial portions of an empirical research project—data cleaning, estimation, table and figure production, and drafting. A central observation of the period is that agents can, in principle, autonomously produce a manuscript of contemporary quantitative empirical research from start to finish. The open questions are therefore no longer *whether* an agent can do the work, but *which stages* it should be trusted with, *how* a human stays meaningfully in the loop, and *what* scaffolding keeps the output correct and comprehensible to the researcher who signs their name to it.

This document collects the concepts that answer those questions, using Claude Code—Anthropic's command-line coding agent—as the concrete running example.

2 Claude Code as a research "harness"

The organizing metaphor is the **harness**: not a single tool but a staged environment that structures how a researcher and an agent work together. A harness is less a piece of software than a *staging of one's own cognition*, so that a person can move efficiently through a project while retaining control and understanding.

2.1 The three obligations: production, verification, understanding

A useful way to frame the goal of any research harness is three obligations that must all hold at once:

- **Production** — the analysis actually gets built and run.
- **Verification** — the output is confirmed to be what it claims to be and what the researcher intended.
- **Understanding** — the researcher retains a continuous, current grasp of what has been done and why.

Agentic speed threatens the second and third obligations: it is easy to generate correct-looking results faster than one can check or comprehend them—the "drinking from a firehose" problem. A good harness deliberately slows the loop where verification and understanding demand it.

2.2 Design principles over ad-hoc prompting

The harness is built around explicit *design principles* for the research domain (in Cunningham's case, causal inference), often encoded as **checklists**. Checklists discipline the mind, make each analytical step inspectable, and give the agent a stable specification to work against rather than a moving target of one-off instructions.

3 Core building blocks

3.1 Hooks

Hooks are user-defined rules that fire automatically at defined points in the agent's workflow, enforcing invariants the researcher cares about without having to restate them each session. A canonical example from empirical work is the rule *"don't edit raw data"*: a hook can guard the raw-data directory so that no agent action mutates the original inputs, forcing all transformations into reproducible, downstream scripts. Hooks turn good practice into infrastructure rather than intention.

3.2 Connectors and the Model Context Protocol (MCP)

Connectors extend an agent beyond the local machine into external and cloud services. Two capabilities they unlock are especially relevant:

- using the agent from the **phone app**, not only the desktop command line; and
- reaching **cloud services** rather than only local files.

Connectors are built on **MCP (Model Context Protocol)**, an open standard that lets AI applications connect securely to external data sources and tools to read real-time information and take actions—effectively the "piping" between the model and outside services. The connector directory functions like an app store, with thousands of entries; some are first-party (vendor-vetted) and some are labeled **Community** (user-created, only lightly reviewed—"buyer beware").

Illustrative categories.

- *Code* — hundreds of developer-oriented connectors (e.g. "Inkbox," which gives an assistant a communication identity: an email address, phone number, and SMS/iMessage capability).
- *Financial services* — Stripe and PayPal (micropayments), QuickBooks (accounting), Gusto (payroll), Dropbox (files); direct bank API access is conspicuously rare, plausibly for security reasons.

For research specifically, the payoff of connectors is still emerging: the expectation is that some cloud-only resources will become reachable *efficiently only through a connector*, making them instrumental to future workflows even if the immediate wins are in productivity and small-business operations rather than estimation.

3.3 Memory: fighting the residue problem

A recurring difficulty is that agents (and their users) lose the thread of prior work across sessions—captured by the maxim *"memory is the residue of thought."* If thinking is not deliberately externalized, little residue remains, and continuity is lost. Practical responses include persistent project notes, checklists tied to the analysis stage, and—at the interface level— making artifacts re-inspectable (for example, figures rendered as scrollable, "flippable cards") so the researcher can quickly reconstruct what has been done.

3.4 The human in the loop

Where and how the human sits in the loop is treated as a first-order design question rather than an afterthought. The concern is not only accuracy but the researcher's own *sight*—the loss of the direct, visual contact with code and results that a text editor once provided. This motivates **visual harnesses**: dashboards (and, aspirationally, spatial/AR interfaces such as the Apple Vision Pro) that re-embody the work, stage it, and keep the human an active, seeing participant rather than a rubber stamp.

4 Which stages of research to delegate

Because full end-to-end automation is now technically feasible, the responsible question becomes *selective delegation*: matching each stage of the research production function to the degree of autonomy it can safely bear.

1. **Data ingestion & cleaning** — high leverage, but must be guarded (see the "don't edit raw data" hook) and kept reproducible.

2. **Estimation & specification** — benefits from agent speed, but is exactly where domain design principles and verification must bite hardest (see the covariate discussion below).
3. **Tables, figures & reproducibility artifacts** — strong fit for automation, provided outputs remain inspectable.
4. **Interpretation & writing** — assistive, but the researcher must retain authorship and understanding.

The governing principle is that autonomy should expand only as far as the harness can still guarantee verification and understanding.

5 Domain application: causal inference

The clearest test of these ideas is a substantive methodological domain. In recent work on **covariate inclusion** in difference-in-differences, Cunningham distinguishes "rungs" of how and when covariates enter estimators such as **Callaway and Sant'Anna**. The relevance to AI-assisted research is direct:

- Such choices are *design decisions*, not mechanical steps, so they are precisely what a checklist-driven harness should surface for human judgment rather than silently automate.
- They show why an agent's fluent, plausible output is insufficient: without an explicit specification of *which* covariates and *when*, the "correct-looking" result may embed an unjustified modeling choice.

The broader point: the value of the harness is that it forces the estimator's assumptions to the surface, where a knowledgeable researcher—not the agent—owns them.

6 Economic context: solopreneurs and productivity

The same connector-and-agent stack that supports research also lowers the cost of running a business, giving rise to the **"solopreneur"**: a firm with a handful of employees, sometimes just one. Payment-processor data (e.g. Stripe's economics team) indicate rapid growth in solo founding since roughly the second quarter of 2025, with a nontrivial share of such firms reaching \$1M–\$10M in annual revenue with minimal headcount.

Framed in growth terms, if Y is output, N is labor, and Pop is population, per-capita income can be written schematically as

$$\frac{Y}{\text{Pop}} = f\left(\frac{Y}{N}, \dots\right),$$

so that gains in labor productivity (Y/N) plausibly propagate—through market mechanisms and labor income—into per-capita income (Y/Pop). The suggestion is that agentic tools may raise individual productivity enough to matter at this aggregate level, feeding both the "wealth of nations" (growth) and its distribution (the "wages of nations").

7 Open challenges

- **Verification at speed** — keeping checking and comprehension from falling behind generation.
- **Trust and vetting** — Community connectors are only lightly reviewed; sensitive integrations (e.g. banking) remain gated, partly over data-breach risk.
- **Loss of "sight"** — restoring the researcher's direct, visual relationship to code and results through better (possibly spatial) interfaces.

- **Fit for research vs. operations** — connectors' payoff for empirical estimation is still less proven than their payoff for business/productivity tasks.

8 Conclusion

Claude Code exemplifies a shift from AI-as-assistant to AI-as-agent in research. The durable concepts are not the individual features but the *harness* philosophy that surrounds them: hooks that encode good practice as infrastructure, connectors/MCP that extend reach into cloud services, deliberate memory and visual design that preserve understanding, and an explicit contract of production, verification, and understanding. Delegation should expand only as fast as that contract can be honored—and in a methodological field such as causal inference, the design decisions that define an estimator must remain the researcher's own.

Sources and further reading

Primary source: *Scott's Mixtape Substack* by Scott Cunningham, https://causalinf.substack.com/. Concepts above draw on posts from August–September 2026, including:

- "Connectors and My Harness Idea" (https://causalinf.substack.com/p/connectors-and-my-harness-i 9 Sep 2026.
- "Four Rungs of Covariate Inclusion in Callaway and Sant'Anna," 7 Sep 2026.
- "Where and how will the human be in the loop?," 4 Sep 2026.
- "Which covariates? When to include them?," 3 Sep 2026.
- "Some thoughts about AI in practice," 2 Sep 2026.
- "How and why I am using hooks (part 1)," 24 Aug 2026.
- " 'Memory is the residue of thought' and fixing that when working with AI agents," 20 Aug 2026.
- "Which stages of research production should we let AI Agents loose," 14 Aug 2026.

Related: *Causal Inference: The Remix* (Yale University Press, 2026); Model Context Protocol (https://modelcontextprotocol.io); Stripe on solo founders.